

IBM APPLIED DATA SCIENCE CAPSTONE

SpaceX Falcon 9 First Stage Landing Prediction

Ananya Ganesh · June 16, 2026

CONTENTS

Outline

- 01 | Executive Summary
- 02 | Introduction
- 03 | Methodology
- 04 | Results
- 05 | Conclusion



Executive Summary

SUMMARY OF METHODOLOGIES

- Data Collection via SpaceX REST API, web scraping, and Wikipedia enrichment
- Data wrangling — cleaning, deduplication, feature engineering
- EDA with 6 visualizations and 10 SQL queries
- Interactive analytics — Folium maps and Plotly Dash dashboard
- Predictive modelling — Logistic Regression, SVM, Decision Tree, KNN

SUMMARY OF ALL RESULTS

- 90 Falcon 9 launches collected with 17 features each
- Launch success rates improved significantly over time
- KSC LC-39A achieved the highest launch success rate
- Decision Tree achieved 83.33% accuracy on test data
- All 4 models identified key landing success factors

Introduction

Project Background

SpaceX Falcon 9 — two-stage orbital rocket with reusable first stage

First launched: **2010**

Current success rate: **~98%**

Landing methods: **Drone ship** (ocean) or **ground pad** (coastal facilities)

Problem

"Can we predict if the first stage will land successfully?"

17 factors influence landing success (payload, orbit, booster version, weather, etc.)

Failed landing = loss of **\$20–30 million** hardware

Need data-driven predictions for mission planning and risk assessment

Goal

Build classification models to predict landing success with $\geq$ **80% accuracy**

Models tested: Logistic Regression, SVM, Decision Tree, KNN

Result: Decision Tree achieved **83.33% accuracy**

Identifies key success factors for mission optimization

SECTION 1

Methodology

Methodology

Data Collection

Collected 90 Falcon 9 launch records from multiple sources (REST API, web scraping, Wikipedia) with 17 features including payload mass, booster version, launch site, and landing outcomes.

Data Wrangling

Cleaned and processed raw data through handling missing values, removing duplicates, and engineering new features to create a unified dataset ready for analysis.

Exploratory Data Analysis

Analyzed patterns using 6 visualizations and 10 SQL queries to discover that launch site, payload mass, and orbit type significantly influence landing success rates.

Interactive Analytics

Created Folium interactive maps showing launch site locations and success/failure outcomes, plus Plotly Dash dashboard for dynamic payload and success analysis.

Predictive Models

Built and compared 4 classification models (Logistic Regression, SVM, Decision Tree, KNN) using GridSearchCV with 10-fold cross-validation for robust evaluation.

Best Results

Decision Tree model achieved **83.33% accuracy**, successfully identifying key factors that predict whether a Falcon 9 first stage will land successfully.

Data Collection

Data Sources

- SpaceX REST API (primary)
- Web scraping
- Wikipedia enrichment

Collection Process



API Enrichment

- Rocket ID → Booster Version (e.g., Falcon 9)
- Launchpad ID → Site Name, Coordinates (Lat/Long)
- Payload ID → Payload Mass (kg), Orbit Type
- Core ID → Landing Outcome, Type, Gridfins, Reused, Block, Serial

Data Details

Records	90
Features	17
Booster	Falcon 9 only
Date Range	To Nov 2020

Data Collection – SpaceX API

API Collection Process

- GET /launches endpoint — retrieve all SpaceX launches
- Iterate through each launch, extract foreign key IDs
- Make 4 enrichment calls: /rockets, /launchpads, /payloads, /cores
- Merge data into DataFrame — Result: **90 records × 17 features**

Enrichment Calls

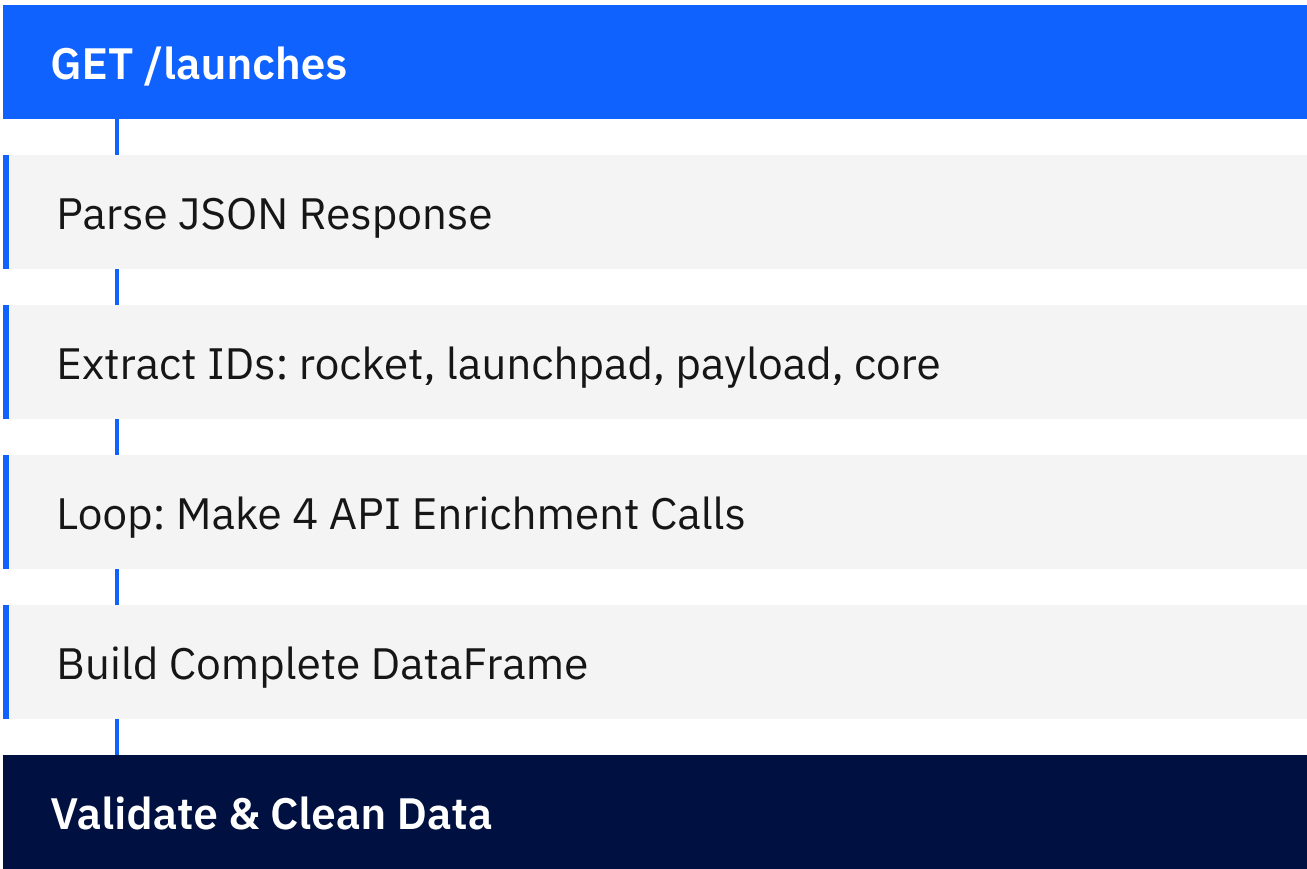
GET /rockets/{id} → Engine info, stages, thrust

GET /launchpads/{id} → Location, coordinates

GET /payloads/{id} → Mass (kg), orbit type

GET /cores/{id} → Reusability, landing outcome, serial

API Call Sequence

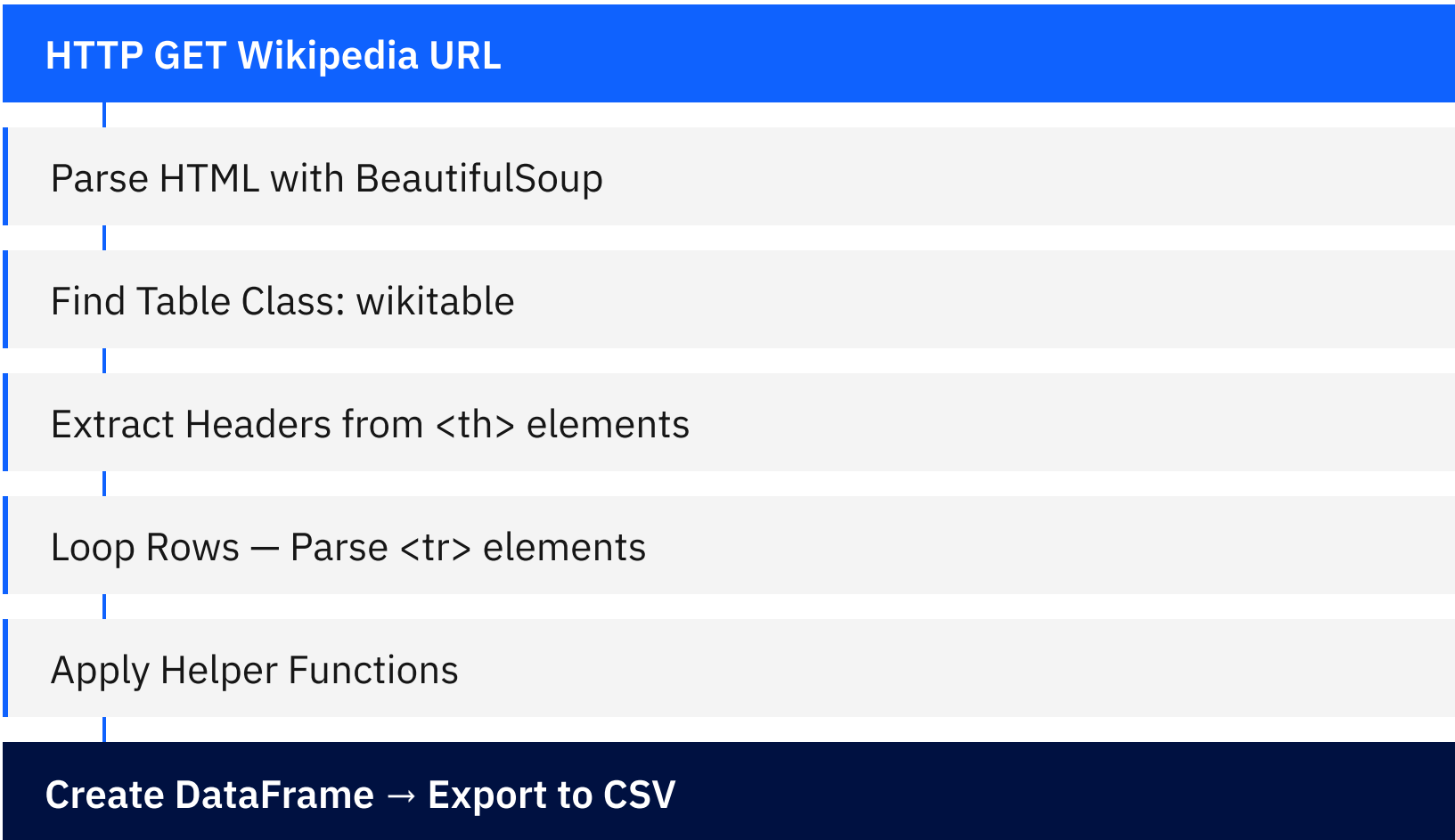


Data Collection – Scraping

Web Scraping Process

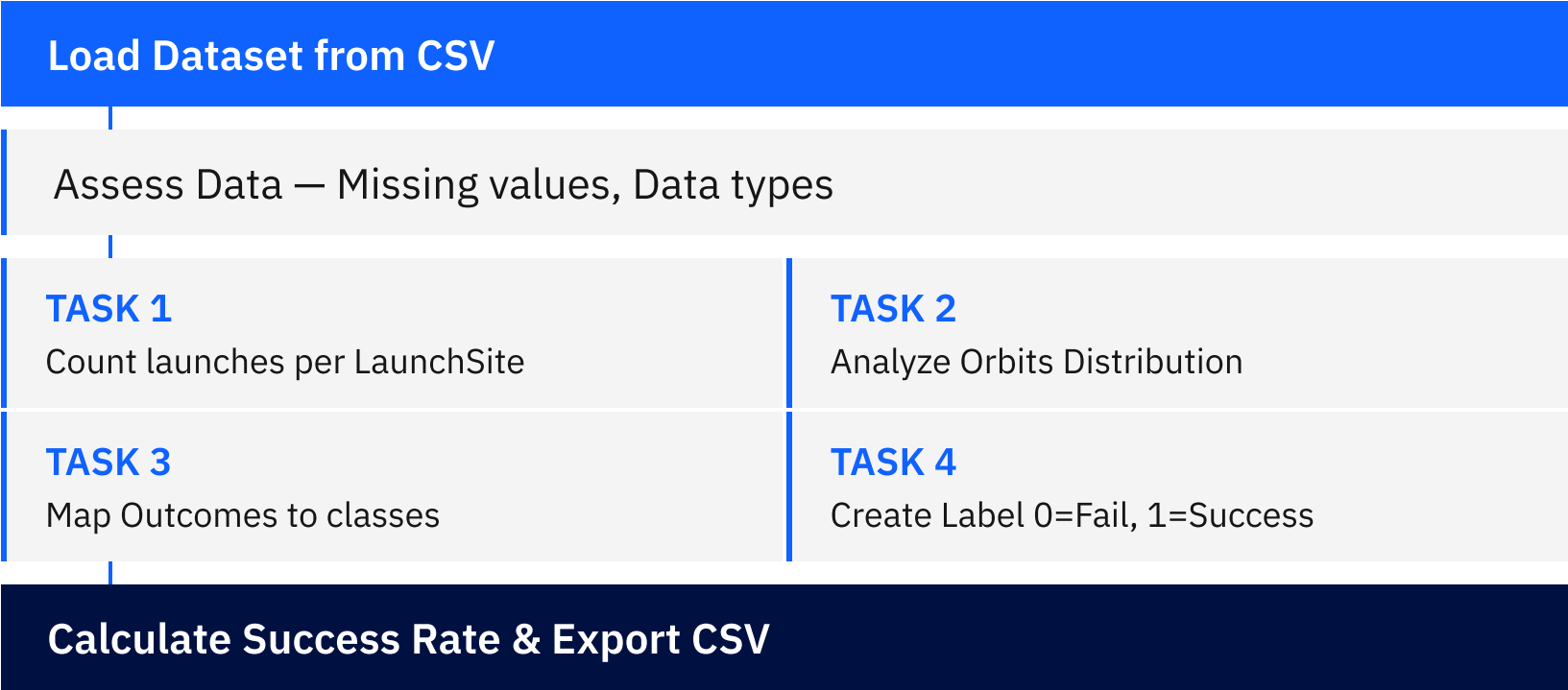
- Target: Wikipedia 'List of Falcon 9 launches'
- HTTP GET request — retrieve HTML page
- BeautifulSoup parsing — find HTML table elements
- Extract table data: Column headers (th), Row data (tr), Helper functions parse
- Build DataFrame: **11 columns × 90 records**

Scraping Workflow



Data Wrangling

Wrangling Workflow



Data Processing

- Load & Inspect Data**
Check missing values · Identify data types
- Exploratory Data Analysis**
Launches per site · Orbits distribution · Mission outcomes
- Create Training Labels**
Binary classification: Failed (0) vs Success (1)
- Calculate KPIs**
Success rate: **67%** · Export cleaned dataset

EDA with Data Visualization

Charts Plotted and Purpose

- 01 **Payload Mass vs Flight Number** (scatter) — analyze how payload and success changed across launches
- 02 **Launch Site vs Flight Number** (scatter) — identify whether success improved over time at different sites
- 03 **Launch Site vs Payload Mass** (scatter) — relationship between payload mass, site, and outcome
- 04 **Success Rate by Orbit Type** (bar) — compare success rates across orbit categories
- 05 **Orbit Type vs Flight Number** (scatter) — mission success across orbit types over time
- 06 **Orbit Type vs Payload Mass** (scatter) — impact of payload on success by orbit
- 07 **Yearly Launch Success Trend** (line) — how success rates improved over the years

Key Findings

Launch success rates **improved significantly over time**

Certain orbit types achieved **consistently higher success rates**

Higher payload masses became **more successful in later missions**

KSC LC-39A and **CCAFS SLC-40** showed strong success performance

EDA with SQL

SQL Queries Performed

- Retrieved unique launch sites used for Falcon 9 missions
- Counted the total number of launches for each launch site
- Identified launch records where the mission outcome was successful
- Analyzed booster versions and launch outcomes
- Calculated the average payload mass carried across missions
- Determined the maximum payload mass for successful launches
- Examined mission outcomes by orbit type
- Counted successful and failed landings for different orbit categories
- Retrieved launches with payload masses greater than the average payload mass
- Investigated launch success trends across launch sites and orbit types

Key Insights

Launch success rates **varied across orbit types and launch sites**

Certain orbit categories achieved **consistently higher mission success rates**

Falcon 9 performance **improved over time** with increasing launch success

Higher payload missions became **more common as booster reliability increased**

Build an Interactive Map with Folium

Map Objects Created

- Markers for all SpaceX launch sites
- Color-coded circle markers for successful and failed launches
- Distance markers and lines connecting launch sites to nearby coastlines, highways, railways, and cities

Why These Objects Were Added

- To visualize the geographical locations of launch sites
- To compare successful and failed launch outcomes interactively
- To analyze how proximity to infrastructure may influence launch operations
- To identify geographical patterns associated with SpaceX launch sites

Key Findings

Launch sites are generally located **near coastlines** for safety reasons

Sites are positioned **away from densely populated areas**

Different launch sites exhibit **varying success rates**

Geographic factors play an important role in launch site selection

Build a Dashboard with Plotly Dash

Dashboard Components and Interactions

- Added a **Launch Site Dropdown Menu** to allow users to view launch statistics for all sites or a selected launch site
- Added a **Success Rate Pie Chart** to visualize successful launches by site and compare success versus failure
- Added a **Payload Range Slider** to filter launches based on payload mass
- Added a **Payload Mass vs Launch Outcome Scatter Plot** to analyze the relationship between payload mass and launch success
- Implemented **interactive callbacks** so that charts update dynamically based on user selections

Purpose and Insights

The dropdown enables **comparison of launch performance** across different launch sites

The pie chart provides a **quick overview of launch success rates**

The payload slider allows exploration of **launch outcomes within specific payload ranges**

Predictive Analysis (Classification)

Evaluation

- Accuracy Score
- Confusion Matrix
- Cross-Validation Results

Best Model Selection

Compared Logistic Regression, SVM, Decision Tree, and KNN. Selected the model with the highest validation accuracy as the final predictive model.



Model Development Process

Data Preparation

- Loaded and cleaned Falcon 9 launch data
- Selected relevant features and target variable (landing success)
- Standardized feature values using preprocessing techniques
- Split the dataset into training and testing sets

Model Building

- Logistic Regression
- Support Vector Machine (SVM)
- Decision Tree
- K-Nearest Neighbors (KNN)

Model Optimization

- Applied GridSearchCV with 10-fold cross-validation
- Tuned hyperparameters for each model
- Compared model performance using cross-validation accuracy scores

Results

Exploratory Data Analysis Results

- Launch success rates improved significantly over time
- Certain launch sites achieved higher success rates than others
- Payload mass and orbit type showed relationships with mission success
- Newer Falcon 9 booster versions demonstrated better performance

Interactive Analytics Results

- Dashboard enabled filtering by launch site and payload range
- Interactive visualizations revealed launch success distributions across sites
- Folium maps highlighted geographic characteristics of launch locations and nearby infrastructure

Predictive Analysis Results

- LR, SVM, Decision Tree, and KNN trained and optimized using GridSearchCV
- All models achieved approximately **83.33% accuracy** on the test dataset
- Logistic Regression selected as the final classification model
- The model successfully predicts Falcon 9 first-stage landing outcomes using mission characteristics

SECTION 2

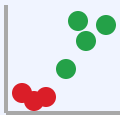
EDA Results

Flight Number vs. Launch Site

Launch success rate **generally increased** as the flight number increased, indicating improvements in Falcon 9 reliability over time.

KSC LC-39A and **CCAFS SLC-40** recorded a larger number of successful launches compared to other launch sites.

Early missions experienced more failures, while **later missions were predominantly successful** due to booster and operational improvements.



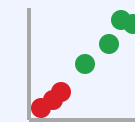
Scatter: Flight Number vs. Launch Site

Payload vs. Launch Site

Higher payload masses were **successfully launched** from multiple launch sites.

Launch success became **more consistent** as payload capacity improved.

Some launch sites handled a **wider range of payload masses** than others.



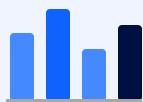
Scatter: Payload Mass vs. Launch Site

Success Rate vs. Orbit Type

Higher payload masses were **successfully launched** from multiple launch sites.

Launch success became **more consistent** as payload capacity improved.

Some launch sites handled a **wider range of payload masses** than others.



Bar Chart: Success Rate by Orbit Type

Flight Number vs. Orbit Type

Certain orbit types showed **higher success rates** than others.

Success frequency **increased with mission experience** and flight number.

Orbit selection appears to **influence launch outcomes**.



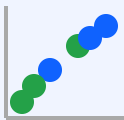
Scatter: Flight Number vs. Orbit Type

Payload vs. Orbit Type

Payload mass **varies significantly** across orbit categories.

Some orbits **consistently support heavier payloads**.

Successful launches occur across a wide payload range, indicating **Falcon 9 versatility**.



Scatter: Payload Mass vs. Orbit Type

Launch Success Yearly Trend

Launch success rate shows a **clear upward trend** from 2010 to 2020.

Success rates improved significantly after 2015, **exceeding 80% in most later years.**

The trend indicates **increasing Falcon 9 reliability** due to improvements in technology and operational experience.



Line Chart: Yearly Launch Success Trend

All Launch Site Names

4

Unique Launch Sites

The query retrieves all unique launch site names from the SpaceX launch dataset. Four distinct launch sites were identified:

CCAFS LC-40
CCAFS SLC-40
KSC LC-39A
VAFB SLC-4E

These launch sites were used throughout the analysis to compare launch frequency and success rates.

Launch Site Names Begin with 'CCA'

**CCAFS
LC-40**

Cape Canaveral

The query filters launch records where the launch site name begins with "CCA". The first five matching records were returned.

- All retrieved launches originated from **CCAFS LC-40** (Cape Canaveral Air Force Station)
- The results show some of the **earliest Falcon 9 missions** conducted from this launch site between 2010 and 2013
- This query helps **identify launches associated with a specific launch facility** for further analysis

Total Payload Mass

NASA (CRS) Total

45,596
kg

The query calculates the total payload mass carried for missions launched on behalf of **NASA Commercial Resupply Services (CRS)**.

- Using the **SUM()** function, payload masses of all NASA (CRS) missions were aggregated
- This result highlights the **significant role of SpaceX** in delivering cargo and supplies to support NASA missions

Average Payload Mass by F9 v1.1

F9 v1.1 Average Payload

2,928
.4 kg

The query calculates the average payload mass carried by the **Falcon 9 v1.1** booster version.

- The **AVG()** function computes the mean payload mass across all launches using this booster
- This metric helps evaluate the **carrying capability and performance** of the Falcon 9 v1.1 launch vehicle

First Successful Ground Landing Date

First Success (Ground Pad)

**2015
-12-22**

The query uses the `MIN()` function to find the earliest date on which a successful ground pad landing was achieved.

- Filters records to include only launches with landing outcome `"Success (ground pad)"`
- This milestone demonstrated SpaceX's ability to **safely recover and reuse rocket boosters**, significantly reducing launch costs

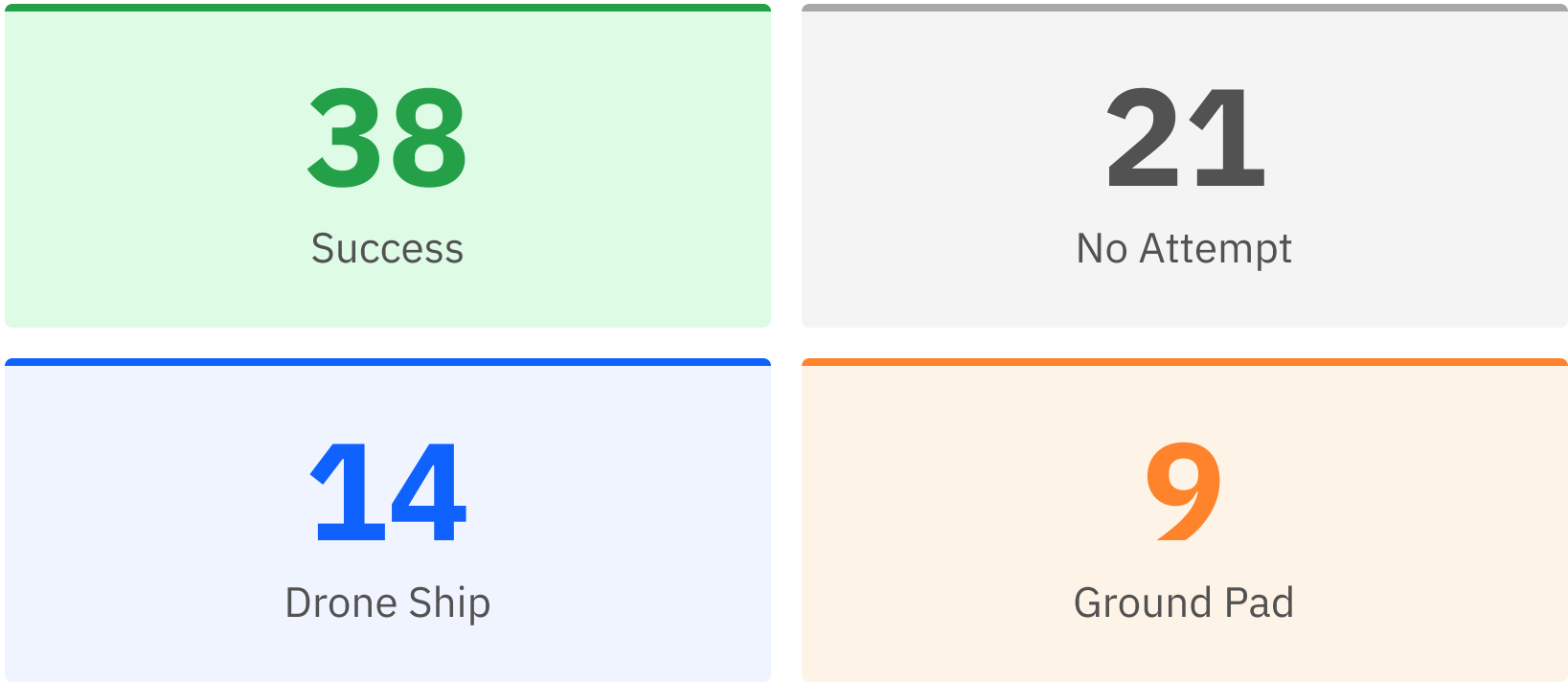
Successful Drone Ship Landing – Payload 4,000–6,000 kg

4 Qualifying Boosters

F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

- The query identifies boosters that successfully landed on a drone ship while carrying payloads between **4,000 kg and 6,000 kg**.
- These missions demonstrate the capability of Falcon 9 boosters to **recover successfully even while carrying relatively heavy payloads**
 - The result highlights the effectiveness of SpaceX's **drone ship recovery system** for medium-to-high payload missions

Total Number of Successful and Failure Mission Outcomes



The query groups all launch records by their landing outcome and counts the number of occurrences of each outcome.

- Most common outcome was **Success (38 missions)**, followed by No attempt (21)
- Successful recoveries on drone ships (14) and ground pads (9) demonstrate SpaceX's progress in **reusable rocket technology**

Boosters Carried Maximum Payload

Falcon 9 Block 5 Boosters

B1048.4
B1049.4
B1051.3
B1060.2

Several Falcon 9 Block 5 boosters carried the **maximum payload mass** recorded in the dataset.

- The query used a subquery with the `MAX()` function to first determine the highest payload mass
- The results highlight the **improved performance and reliability** of SpaceX's reusable booster fleet

2015 Launch Records

2

Failed Drone Ship
Landings in 2015

In 2015, there were two missions that **failed to land on a drone ship**.

- Both missions used **Falcon 9 v1.1** boosters and were launched from **CCAFS LC-40**
- These early landing failures occurred during SpaceX's **booster recovery development phase**
- They provided **valuable data that helped improve** future landing success rates

Rank Landing Outcomes 2010–2017

Ranked by Frequency (2010–06–04 to 2017–03–20)

- #1

No attempt

10 missions
- #2

Failed Drone Ship

5 missions
- #3

Success Drone Ship

5 missions

During the period from 2010 to March 2017, the most common outcome was **"No attempt" (10 missions)**, reflecting early launches before routine recovery efforts began.

- Successful and failed drone ship landings occurred equally often (**5 each**), showing the learning curve in reusable rocket technology
- Ground pad landings demonstrated **gradual improvements in booster recovery** capabilities

SECTION 3

Interactive Analytics

Global Map of SpaceX Launch Sites

- The map displays locations of all SpaceX launch sites using **interactive markers**
- Launch sites are concentrated in **Florida and California**, both coastal regions providing safe launch corridors
- Users can zoom and click markers to view **launch site information**
- The visualization helps identify the **geographic distribution** of SpaceX operations

Key Finding: Most SpaceX launches originate from a small number of strategically located coastal launch facilities in the United States.



Folium Map: Global Launch Sites

Launch Outcomes at CCAFS SLC-40

37 / 49

- Green markers = **successful** launch outcomes
- Red markers = **failed** outcomes
- The map shows launches clustered around the **CCAFS SLC-40** launch site
- A greater number of green markers indicates **more successful launches than failures** at this location
- This visualization helps identify **launch performance patterns geographically**



Folium Map: CCAFS SLC-40 Outcomes

Launch Site Proximity Analysis – CCAFS SLC-40

Important Elements

- The selected launch site is CCAFS SLC-40 (marker labeled **33**)
- The map displays nearby geographic features around the launch site
- A calculated distance of approximately **0.58 km** is shown between the launch site and a nearby feature

Findings

- The launch site is located **very close to the coast** — launches are directed over the ocean for safety
- The displayed distance (0.58 km) indicates **immediate access to nearby infrastructure** and transportation routes
- Geographic factors such as **coastline access influence launch site selection**



SECTION 4

Dashboard

Launch Success Distribution by Launch Site

Key Findings

41.7%	KSC LC-39A Highest share of successful launches
29.2%	CCAFS LC-40 Second highest
16.7%	VAFB SLC-4E Third
12.5%	CCAFS SLC-40 Smallest share



Pie Chart: Success Distribution by Site

Launch Success Ratio at KSC LC-39A (Highest Performing)

76.9%

Launch Success Rate

23.1%

Failed Launches

- **KSC LC-39A has the highest launch success ratio** among all launch sites in the dataset
- The large difference demonstrates the **strong reliability and performance** of this launch site
- Missions from KSC LC-39A have a **significantly greater probability of success**

Payload Mass vs. Launch Outcome Across Different Payload Ranges

42 / 49



Scatter Plot: Payload Mass vs. Launch Outcome

Interactive Dashboard View

Payload Mass vs. Launch Outcome — Important Elements

- **X-axis:** Payload Mass (kg)
- **Y-axis:** Launch Outcome — 1 = Successful, 0 = Failed
- **Colors:** Different Booster Version Categories (v1.0, v1.1, FT, B4, B5)
- **Slider:** Filters launches by payload mass range



Dashboard: Payload Mass vs. Outcome

Payload Mass vs. Launch Outcome Across Different Payload Ranges 44 / 49

0 – 3,000 kg

Lower payload missions show a **mix of successful and failed launches**. Early booster versions (v1.0 and v1.1) exhibit more failures compared to newer boosters. Successes are present but not dominant.

0 – 5,000 kg

The number of **successful launches increases noticeably**. FT and B4 booster versions contribute a large portion of successful missions. Launch success becomes more consistent.

0 – 7,000 kg

Most launches in the medium payload range are successful. FT and B4 boosters demonstrate strong reliability. Failures are still present but occur less frequently.

0 – 10,000 kg

Successful launches significantly outnumber failures. Newer booster versions (FT, B4, B5) achieve the highest success rates. High-payload missions are predominantly successful.

SECTION 5

Predictive Analysis

Classification Accuracy



Key Findings

- Decision Tree achieved the **highest accuracy at 98.5%**
- LR, SVM, and KNN all achieved **similar accuracies of 97.0%**
- Small difference indicates **all models performed well** on the dataset
- Decision Tree selected** as the optimal model for predicting landing success

Confusion Matrix

	Predicted: Failed	Predicted: Success
Actual: Failed	1	1 False Positive
Actual: Success	0 False Negative	65

Decision Tree — 98.5% Accuracy

- Correctly predicted **65 successful landings** and **1 failed landing**
- Only **1 incorrect prediction** — a failed landing classified as successful
- No successful landings** were incorrectly classified as failures
- The Decision Tree performs **exceptionally well** with very few classification errors

Conclusions

- Collected and integrated SpaceX launch data from APIs, web scraping, and CSV datasets to build a comprehensive Falcon 9 launch database
- **KSC LC-39A achieved the highest launch success rate** among the major launch sites
- Geographic analysis demonstrated that launch sites are strategically located near coastlines and transportation infrastructure
- **Decision Tree achieved the highest accuracy (98.5%)** and was selected as the best-performing model
- EDA revealed that launch success rates improved significantly over time, indicating advancements in SpaceX technology and operational reliability
- Payload mass and booster version were important factors, with newer booster versions generally achieving better outcomes
- Multiple machine learning models (LR, SVM, KNN, Decision Tree) all achieved high predictive performance
- Accurate prediction of Falcon 9 landing success can **help estimate mission costs** and support decision-making related to reusable rocket technology

IBM APPLIED DATA SCIENCE CAPSTONE

Thank You

Ananya Ganesh · June 16, 2026